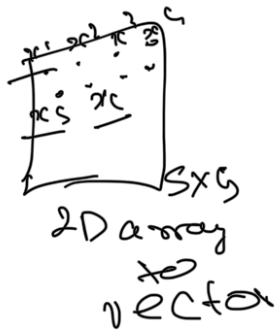
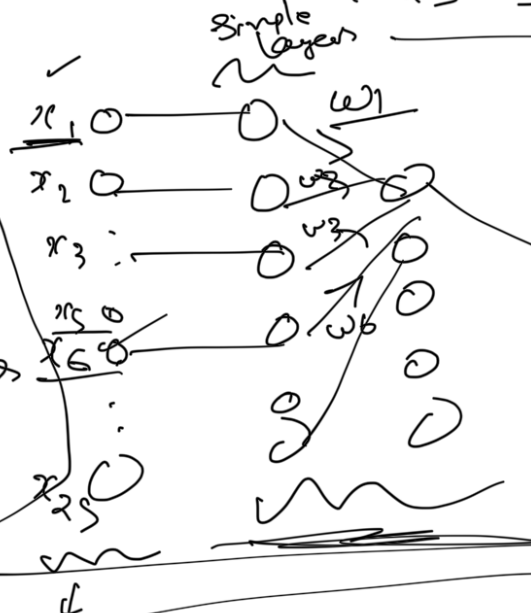


Neural Networks



Aim is to use a feed forward neural network on this 5x5 image

All Architectures
Number of Neurons in output layer = Number of classes



$$\hat{y} = f(x_1, \dots, x_5)$$

$$J = \frac{1}{M} \sum_{i=1}^M (\hat{y}_i - y_{i, \text{true}})^2$$

Number of neurons in input layer = Number of input features $\Rightarrow$ only for feedforward

Differences in image data

Image is always spatially localized.

Smoothly varying information

network should

The neural net.
understand this so as to
learn weights
appropriately

$y = ax + b$
2D problems

$y = ax + by + c$
2D problems

61000
Cenet

~~loss~~ $J = \frac{1}{M} \sum_{i=1}^M (\hat{y}_i - y_i^{true})^2$

$\hat{w} = \min_w J$

$\underbrace{\text{\# of Parameters}}_{\text{Million}}$

Propose
some
structural
or
architectural
changes

Make this
optimization
easier
for
the neural
network.

Simple layer $\Rightarrow \sigma$
 $\vdots$
 σ
 # of neurons

Convolutional layer

Simple layer
 σ
 $\vdots$
 σ
 # of neurons

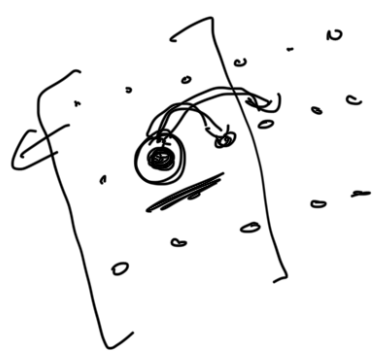
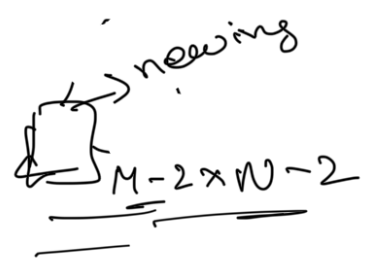
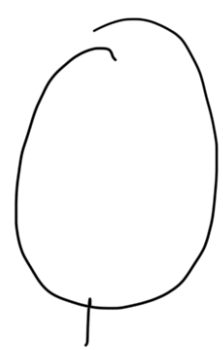
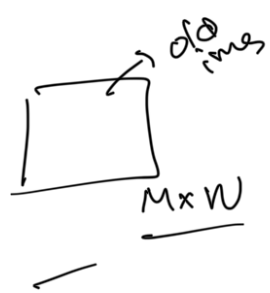
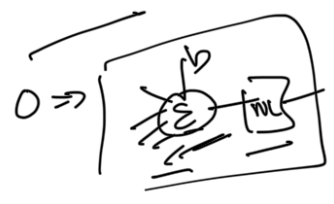
2D array Graphically



input feature maps

output feature maps

kernel (Linear Combiner) (Σ)
 + Non-linearity (ReLU)
 + bias



Kernel $\Rightarrow 3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9$
 spatial filtering

$$\begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix} = h$$

$$\text{3x3}$$

Stride $\Rightarrow 1$

Non-linear $\Rightarrow$ ReLU

Bias term $\Rightarrow$ one for each # of kernel feature

newimg oldimg map

$$\text{newimg}[i,j] = \text{oldimg}[i-1:i+2, j-1:j+2] \times h$$

$$\text{newimg}[i,j] = \text{relu}(\text{newimg}[i,j])$$

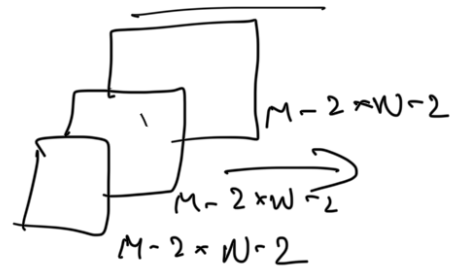
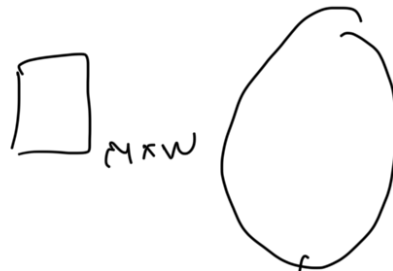
$$\text{newimg}[i,j] = \text{newimg}[i,j] + b$$

Same for
one
output
feature
map.

Can a convolutional layer
give more
than one feature maps
as output

Yes
&
Use
Multiple
Kernels

eg. 3 Kernels



of Parameters
in
this layer

$$= \text{\# of Kernel weights} + \text{\# of bias terms}$$

$$= 3 \text{ Kernels} + 3 \text{ bias terms}$$

$$= 3 \times 3 \times 3 + 3$$

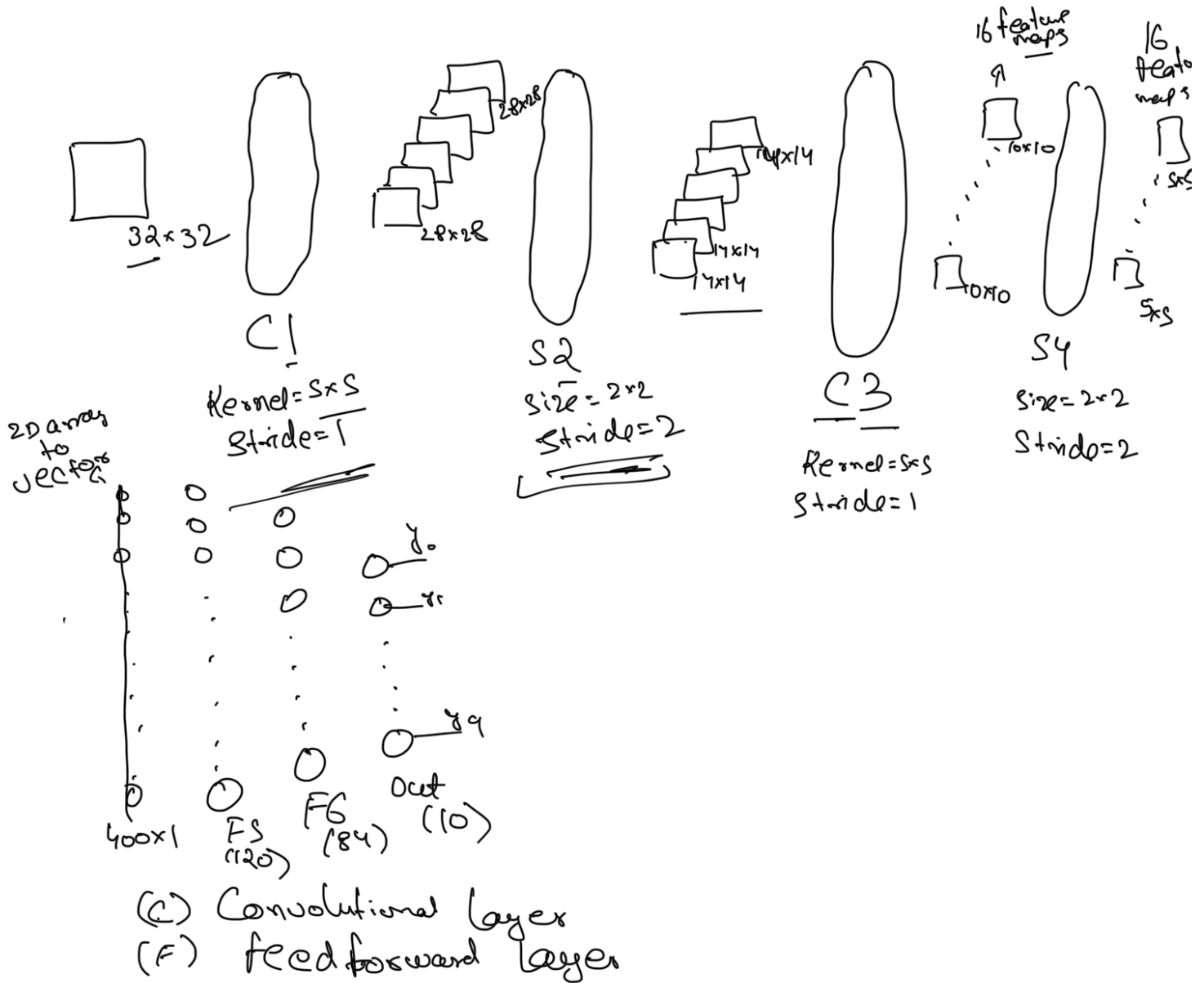
$$= 30$$

16x 5x5
16x25

~~$y = x$
 $y = 0$~~
 $x > 0$
 $x < 0$

LeNet

16

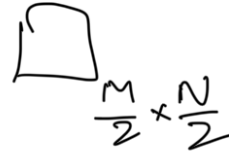


(S) Sub Sampling layer

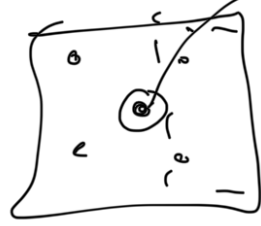
↓
Reduces
the
size
of
input
array.



Stride = 2
Size = 2×2



↓
to get
coarser
information!



replace
4 pixel
with
a single
value
↓
average
value

